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The NANO-2024 Conference was organized by the Institute of Physics of NAS of Ukraine with the participation of the University of Tartu (Estonia), the Uzhhorod National University, University of Turin (Italy) and Pierre and Marie Curie University – Paris 6 (France).

NANO-2024 is the 12th conference in the series of NANO-conferences initiated by the Institute of Physics of NAS of Ukraine in 2012 in the framework of FP7 Nanotwining project. From year to year, they attract more attention and participants. In 2012, the first meeting was held in the format of International Summer School for young scientists «Nanotechnology: from fundamental research to innovations». The 2013 and 2014 conferences were organized in conjunction with the International Summer Schools for young scientists under the same title. In 2013, this event was attended by more than 300 scientists, in 2014-2017, 450 scientists took part and in 2018 it gathered above 650 participants. In 2021 conference was attended by more than 700 scientists from Ukraine, Poland, Italy, Estonia, France, Austria, Germany, Greece, Turkey, USA, Romania, Moldova, Czech Republic, Taiwan, Lithuania, Egypt, India, Algeria, Indonesia and other countries. In 2021 - 2023 the Organizer Committee has received more than 800 application forms from about 25 countries of the world each years.

The NANO-2024 conference brought together leading scientists and young researchers from many countries of the world. This year its topics were as follows: Nanobiotechnology for health-care; Nanochemistry and biotechnology; Nanocomposites and nanomaterials; Nanoobjects microscopy; Nanooptics and photonics; Nanoplasmonics and surface enhanced spectroscopy; Nanoscale physics; Nanostructured surfaces; Physico-chemical nanomaterials science.

This year, the conference is organized with the support of the project "Knowhow Communities for Accelerating RTI Transfer in the Danube Region" (RTIT>>), DRP0200330.

Website of the Nano-2024 conference: http://nano-conference.iop.kiev.ua

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Impact of nanoscale terrace-ledge-kink morphology in high dielectric response of titanium-based dielectric ceramics

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Electrically heterogeneous ceramics based on $CaCu_3Ti_4O_{12}$ (CCTO) have been a topic of interest for researchers for over two decades due to their high permittivity ε' in a wide temperature range. The high permittivity of such materials is important for the development of energy-storing supercapacitors, and sensors of "electronic skin", and critically depends on its microstructure.

This research aimed to understand how nanoscale surface structures on the grains of CCTO-based ceramic materials affect their electrical response.

It was found that the F introduction into CCTO does not lead to the substitution of O^{2-} with F and does not stimulate the formation of Cu^{1+}. Instead, F^- along with the grinding products stimulates the active migration of copper from the bulk of the grain with the appearance of V''_{Cu} and Cu^{3+}, and the formation of CuO-rich nanoscale (up to 200 nm) terrace-ledge-kink morphology and twinning planes. These structures are responsible for the nanoscale barrier layer capacitance component of the dielectric response of CCTO-based ceramics, with p-type conductance by Cu^{3+} ions. Both undoped and F-doped CCTO ceramics are morphologically and chemically heterogeneous. The increase in the fraction of CuO-rich TLK morphologies with the uniform periodicity of domain walls and insulating planes can decrease dielectric loss and provide the highest dielectric response ($\varepsilon'_{1kHz} \sim 6.9\times10^4$) up to frequencies of 10^5 Hz.

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